Research Gaps: Multilingual Document Intelligence RAG

[RESEARCH GAPS IN MULTILINGUAL RAG LITERATURE]
Project: Multilingual Document Intelligence (RAG)
Purpose: This review identifies the research space for a resource-efficient, trustworthy RAG system for private English and Indian-language document archives.

[1. Investigating Language Preference of Multilingual RAG Systems - ACL Findings 2025]
Contribution: Analyses language preference in multilingual retrieval and generation and proposes DKM-RAG.
Gap: It does not focus on private OCR-derived enterprise documents, page-level provenance, or Indian-language deployment. It motivates evaluating answer-language fidelity, Latin-script bias, and code-switching in the proposed system.

[2. Resource-Efficient Multilingual RAG: English-Only and Multilingual Embeddings for Indian Languages - 2026]
Contribution: Uses an English embedding pipeline with output translation and reports resource savings for 22 Indian languages.
Gap: English-only retrieval can lose domain-specific legal or technical meaning. The unresolved question is when direct multilingual retrieval, translation-based retrieval, or a hybrid route is best for private multilingual documents.

[3. M3-RAG: Unified Multimodal and Multilingual RAG - 2025]
Contribution: Integrates multilingual text retrieval, image retrieval, cross-modal alignment, and modality routing.
Gap: It does not establish trustworthy page-cited QA for heterogeneous private PDF, DOCX, and scanned-document collections under limited hardware. Tables and diagrams remain a future extension for the proposed project.

[4. Translation-Based Embedding Strategies for Non-English Documents - 2024]
Contribution: Studies translation-based embedding strategies for non-English RAG.
Gap: Translation may cause terminology drift, additional latency, and cost. It does not provide an adaptive policy identifying queries for which translation improves retrieval over multilingual embeddings.

[5. A Multilingual Intelligent Document Processing System - 2025]
Contribution: Addresses multilingual document processing and question answering.
Gap: A functional multilingual QA pipeline alone does not demonstrate retrieval-method comparisons, source-faithfulness, out-of-scope behaviour, or robust evaluation across document formats and languages.

[6. LegalRAG: A Hybrid RAG System for Multilingual Legal Information Retrieval - 2025]
Contribution: Improves Bangla-English legal-gazette QA using relevance checking and query refinement.
Gap: Evaluation is domain- and collection-specific. The authors identify difficulty with out-of-context questions. This motivates abstention, evidence sufficiency checks, and broader multilingual private-document evaluation.

[7. Hybrid Layered Retrieval and Task-Aware Embeddings for Efficient Persian RAG Systems - 2026]

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Contribution: Uses task-aware query augmentation and layered hybrid dense-plus-BM25 retrieval for Persian QA.
Gap: The design is Persian- and task/FAQ-specific, while latency rises on limited GPUs. The proposed work examines a simpler deployable hybrid retrieval design for Indian languages, varying document type and hardware budget.

[CONSOLIDATED GAP]
No selected study evaluates an end-to-end private-document RAG system that jointly compares direct multilingual, translation-based, and hybrid retrieval for English and Indian languages while measuring citation correctness, answer-language fidelity, OCR robustness, abstention, latency, and cost.

[EVIDENCE-BASED RESEARCH QUESTIONS]
RQ1: Which retrieval strategy gives the best retrieval quality and cost for each language and document type?
RQ2: Does hybrid dense-plus-lexical retrieval and cross-encoder reranking improve page-cited answer faithfulness?
RQ3: Can evidence-sufficiency checks reduce unsupported answers without unacceptable latency?
RQ4: How consistently does the system answer in the requested language?